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Fair Decisions in Machine Learning

How can a prediction system give people a fair chance?

Based on Equality of Opportunity in Supervised Learning

Moritz Hardt • Eric Price • Nathan Srebro • 2016

Source: Hardt, Price & Srebro (2016), arXiv:1610.02413v1

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What makes a loan decision fair?

A bank uses a computer model to approve or reject loans.

Same rule

Everyone needs the same credit score.

Same total share

Each group gets the same overall approval rate.

Same chance

People who would repay get the same approval rate.

Discuss: Would these three rules always make the same choices?

Paper: §§1–2. Discussion framing, not a claim that one rule fits every setting.

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A score is not a decision

The model predicts. The bank chooses how to use that prediction.

Applicant information

Income, debt and past payments

→

Credit score

An estimate of repayment ability

→

Loan decision

Apply a cutoff: approve or reject

Past examples help train and check the system.
The bank does not know a new applicant’s future repayment result.

Paper: §§1, 2.2 and 4. A cutoff is the score needed for approval.

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Two mistakes, two kinds of harm

	Would repay	Would not repay
Approve loan	**Correct approval** A good borrower gets a loan.	**Wrong approval** The bank takes a repayment loss.
Reject loan	**Wrong rejection** A good borrower loses a chance.	**Correct rejection** The bank avoids a repayment loss.

Paper: §§2 and 4. Wrong rejection = false negative; wrong approval = false positive.

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Two familiar rules are not enough

“Do not use the group label.”

Other information may reveal the group. Location can act as a substitute.

Location → May reveal group identity

“Approve the same share.”

Equal totals do not show who gets approved or who is wrongly rejected.

Same total approvals
≠ same chance for good borrowers

Paper: §1. These are the authors’ critiques; the right fairness rule remains context-dependent.

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Same totals can hide unequal chances

Each group has 10 people who would repay and 10 who would not.

Group A

8 of 10 who would repay: approved

2 of 10 who would not: approved

Group B

6 of 10 who would repay: approved

4 of 10 who would not: approved

Both approve **10 out of 20** people.
But do good borrowers have the same chance?

Invented teaching example, not paper data. Imagine we know who would repay.

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Equal opportunity: protect good borrowers

Among people who would repay, match the approval rate across groups.

8 out of 10

Group A • good borrowers approved

6 out of 10

Group B • good borrowers approved

The example fails this check: the rates are different.
The rule does not choose the shared rate or control wrong approvals.

Paper: §2.1. Numbers are the invented teaching example. Good-borrower approval rate = true positive rate.

Equalized odds: check both mistakes

Match approval rates separately for each type of borrower.

Who are we checking?	Group A	Group B	Rule
People who would repay	8 of 10	6 of 10	Rates must match
People who would not repay	2 of 10	4 of 10	Rates must match

This is stricter than equal opportunity. Our example fails both checks.

Paper: §2. Invented example. Equalized odds matches conditional rates, not necessarily overall accuracy.

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Three rules ask three different questions

Same total share

Do all groups have the same overall approval rate?

Called demographic parity

Equal opportunity

Do people who would repay have the same approval rate across groups?

Equalized odds

Do approval rates match for both borrower types, checked separately?

Choose the rule by asking: which opportunity and which harm matter here?

Paper: §§1–3. Different fairness goals need not agree.

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Keep the model. Change the decision rule.

1. Check past scores, groups and outcomes

↓

2. Find decision rules that meet the check

↓

3. Choose the rule with the lowest error cost

Called “post-processing”

Keep the original scores.

Different groups may need different approval cutoffs.

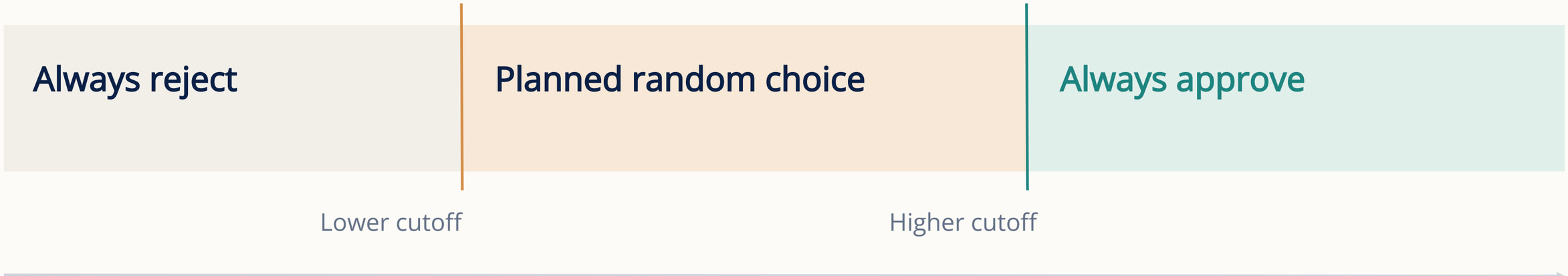
The decision uses score and group, not a new applicant’s future outcome.

Paper: §4. Error cost can give different weights to wrong approvals and wrong rejections.

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Why might the method use a lottery?

Some equalized-odds targets cannot be reached with fixed cutoffs alone.



The method sets the chance in advance to match rates over many decisions—not to guarantee each person’s outcome.

Paper: §4.2. Diagram is conceptual; cutoff locations do not represent measured scores.

What does the theory promise?

- An ideal repayment-probability score keeps all relevant prediction information.
- That score plus group identity is enough to reach the best decision rule under these fairness checks.
- A nearly ideal score needs to be close group by group, not just good on average.

“Best fair” is not:

Perfect prediction

No cost
Any ordinary model

Fairness rules may require accepting more mistakes. Better information can improve the options.

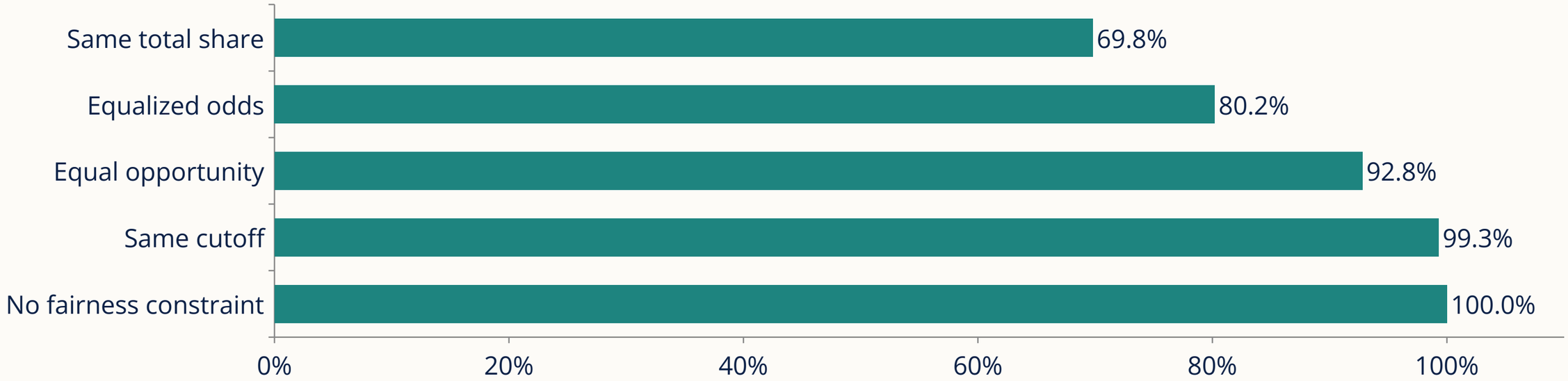
Paper: §5, Proposition 5.2, Corollary 5.3 and Theorem 5.6.

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A credit-score case shows the trade-off

The paper compares rules using 301,536 credit records from 2003.

Share of the maximum possible profit under the paper's cost assumptions



One dataset and one cost setting—not a universal ranking or evidence from a policy trial.

Paper: §7, pp.16–20. TransUnion TransRisk data; four racial groups; any-account default label.
Results use an 82% non-default profitability threshold. "Same total share" = demographic parity.

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A fairness check is not the whole story

Check the records

Past results may be biased or missing. Rejected applicants' repayment outcomes may be unknown.

Look within groups

Equal group averages do not prove that every person is treated fairly.

Ask how it works

Different ways of using information can produce the same score, group and outcome statistics.

Numbers can flag concerns. We still need real-world investigation.

Paper: §§6 and 8. Missing outcomes and individual impacts are application checks, not problems solved by the method.

Three things to remember

1 **Equal opportunity:** match approval rates for people who would repay.

2 **Equalized odds:** also match approval rates for people who would not repay.

3 **Change decisions, then keep checking:** passing a rate check is not proof of fairness.

Discuss: Which mistake matters most here? What would you check beyond the numbers?

Read the paper: arxiv.org/abs/1610.02413v1